🏺 Greenware Inspection System – Setup Guide (Team 2)

📌 Overview

This project is a FastAPI-based backend system for the **Greenware Inspection Dashboard**. It runs locally and can also be deployed on a **Raspberry Pi**.

⚙️ 1. Requirements

For PC / Laptop

- Python **3.10 – 3.12** (⚠️ avoid 3.14 for compatibility issues)
- pip (comes with Python)

For Raspberry Pi

- Raspberry Pi OS (recommended: Bullseye or Bookworm)
- Python 3.9+ (pre-installed usually)

📥 2. Clone the Repository

```shell
git clone https://github.com/YOUR-REPO-LINK.git
cd greenware_pi_app/backend
```

🧪 3. Create Virtual Environment

Windows

```shell
python -m venv .venv
.venv\Scripts\activate
```

Linux / Raspberry Pi

```shell
python3 -m venv .venv
source .venv/bin/activate
```

📦 4. Install Dependencies

```shell
pip install --upgrade pip
pip install -r requirements.txt
```

▶️ 5. Run the Server

```shell
python -m uvicorn app.main:app --host 127.0.0.1 --port 8000
--reload
```

🌐 6. Access the System

Open browser:

- Main:

```
None

http://127.0.0.1:8000/
```

- API Docs:

```
None

http://127.0.0.1:8000/docs
```

🍓 7. Raspberry Pi Access (Important)

If running on Raspberry Pi and accessing from another device (phone/laptop):

Run using:

```
Shell
python -m uvicorn app.main:app --host 0.0.0.0 --port 8000
or
python -m uvicorn app.main:app --host 127.0.0.1 --port 8000
--reload
```

Then access using:

```
None
http://<RASPBERRY_PI_IP>:8000
```

Example:

```
None
http://192.168.1.10:8000
```

🛠️ 8. Common Issues & Fixes

❌ Site can't be reached

- Use `127.0.0.1` instead of `localhost`
- Check if port is used:

```shell
netstat -ano | findstr :8000
```

❌ Stuck at "Waiting for application startup"

- Issue is inside:
  - routes
  - config
  - database
- Test with minimal `main.py`

❌ Static files not loading

Make sure you are inside:

```
None
backend/
```

📁 Project Structure (Simplified)

```
None
backend/
│
├── app/
│   ├── main.py
│   ├── api/
│   ├── core/
│   └── static/
│
├── requirements.txt
└── README.md
```

👥 Notes for Team 2

- Always activate `.venv` before running
- Do NOT use Python 3.14 (causes package errors like pydantic-core)
- Use `/docs` to test APIs
- For UI changes, edit files inside:

```
None

app/static/
```

🚀 Optional (Auto-start on Raspberry Pi)

You can run the server automatically on boot using:

```shell
sudo nano /etc/systemd/system/greenware.service
```

Then add:

```
None
[Unit]
Description=Greenware FastAPI App
After=network.target

[Service]
User=pi
WorkingDirectory=/home/pi/greenware_pi_app/backend
ExecStart=/home/pi/greenware_pi_app/backend/.venv/bin/python -m
uvicorn app.main:app --host 0.0.0.0 --port 8000
Restart=always

[Install]
WantedBy=multi-user.target
```

Then:

```
Shell
sudo systemctl daemon-reexec
sudo systemctl enable greenware
sudo systemctl start greenware
```

🚀 Push / Commit from VS Code to GitHub (SAFE METHOD)

✅ 1. Open your project in VS Code

- Open your folder:

```
None

greenware_pi_app
```

✅ 2. Open Source Control

- Click the **Source Control icon** (🔁) on the left sidebar
- Or press:

```
None

Ctrl + Shift + G
```

✅ 3. Stage your files

- Click **“+” (Stage All Changes)**
  OR type a commit message first, then click **Commit**

✅ 4. Write commit message

Example:

```
None
Updated dashboard UI and developer cards
```

Click ✅ **Commit**

⚠️ 5. FIRST TIME PUSH (IMPORTANT FIX)

If your repo already has files in GitHub (README, etc.), do this:

Open Terminal in VS Code:

```
None
Ctrl + `
```

Then run:

```
Shell
git pull origin main --allow-unrelated-histories
```

👉 Fixes the **"rejected (fetch first)" error**

✅ 6. Push to GitHub

Now go back to VS Code:

- Click **"Sync Changes"**
  OR click **"Push"**

🔁 Alternative (Terminal push)

```shell
git push origin main
```

✅ DONE

Refresh your GitHub repo → your files should appear

⚠️ Common Fixes

❌ Error: rejected (fetch first)

✔ Run:

```shell
git pull origin main --allow-unrelated-histories
```

❌ Nothing happens when pushing

✔ Make sure you committed first

❌ Login required

✔ Sign in via GitHub in VS Code
✔ Or use Personal Access Token

💡 Best Practice

Before every push:

```shell
git pull origin main
```